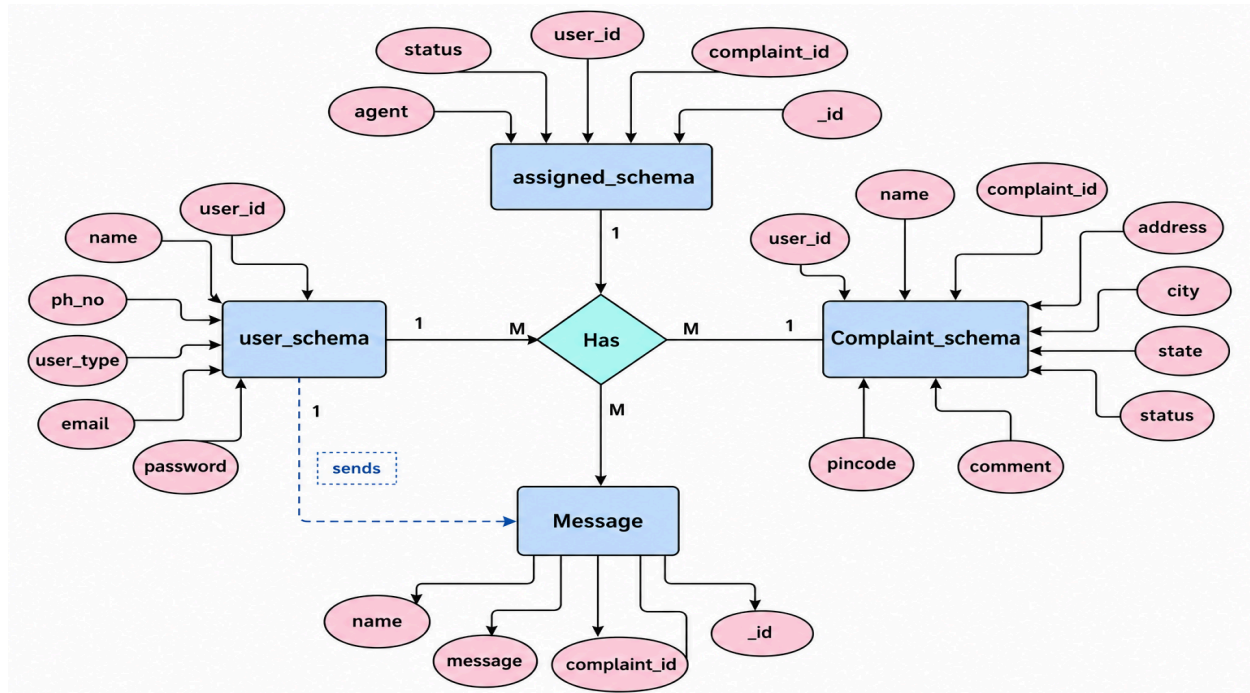




ER Diagram Explanation

The **Entity Relationship (ER) Diagram** represents the database structure of the **Online Complaint Registration System**. It shows the relationship between the main entities of the application: **User**, **Complaint**, **Assigned Complaint**, and **Message**. These entities work together to manage complaint registration, assignment, tracking, and communication between users and agents. The diagram provides a clear understanding of how data flows through the system and how different collections are interconnected.

1. User Schema

The **User Schema** stores the details of registered users who can access the complaint portal.

Attributes

- **user_id** – Unique identifier of the user
- **name** – User's full name
- **email** – Email address used for login
- **password** – Encrypted password
- **ph_no** – Contact number
- **user_type** – Role of the user (Citizen/Admin/Agent)

Purpose

This entity manages user authentication and stores personal information. Every registered user can submit complaints and communicate with assigned agents.

2. Complaint Schema

The **Complaint Schema** stores all complaint information submitted by users.

Attributes

- **complaint_id** – Unique complaint number
- **user_id** – References the user who submitted the complaint
- **name** – Name of complainant
- **address**
- **city**
- **state**
- **pincode**
- **comment** – Complaint description
- **status** – Current complaint status

Purpose

Each complaint belongs to one registered user. The complaint status changes throughout its lifecycle, such as:

- Pending
 - Assigned
 - In Progress
 - Resolved
-

3. Assigned Schema

The **Assigned Schema** records complaint assignments made to agents.

Attributes

- **_id** – Assignment ID
- **complaint_id** – References the complaint
- **user_id** – Complaint owner
- **agent** – Assigned agent
- **status** – Assignment status

Purpose

After an administrator reviews a complaint, it is assigned to an available agent. This schema keeps track of:

- Which complaint is assigned
 - Which agent is responsible
 - Current assignment status
-

4. Message Schema

The **Message Schema** enables communication between users and agents.

Attributes

- **_id** – Message ID
- **name** – Sender's name
- **complaint_id** – Complaint reference
- **message** – Chat message

Purpose

The message collection allows users and agents to communicate regarding a complaint. Every message is linked to a particular complaint using the **complaint_id**.

Relationships Between Entities

User → Complaint (One-to-Many)

- One user can register multiple complaints.
- Each complaint belongs to only one user.

Relationship: 1 : M

Complaint → Assigned Schema (One-to-One)

- Every complaint can be assigned to one agent.

- The assignment record stores the assigned agent and current status.

Relationship: 1 : 1

Complaint → Message (One-to-Many)

- One complaint may contain multiple chat messages.
- Messages help users and agents discuss complaint progress.

Relationship: 1 : M

User → Message

- Users can send multiple messages.
 - Messages are linked to complaints for easy tracking.
-

Conclusion

The ER diagram provides a structured representation of the **Online Complaint Registration System** by defining the relationships between **User**, **Complaint**, **Assigned**, and **Message** entities. It ensures proper organization of complaint data, efficient assignment of complaints to agents, secure user management, and seamless communication between users and agents. This database design supports all major functionalities of the system, including user registration, complaint submission, complaint tracking, assignment management, status updates, and real-time messaging, making the application efficient, scalable, and easy to maintain.